Bringing Interpretability to Neural Audio Codecs

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CSCI 682 - Paper Discussion

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Why Interpretability for Audio Codecs?

- In NLP, tokens map to known linguistic units
  - We can probe what BERT encodes at each layer
- Neural audio codecs convert speech waveforms into discrete token sequences (the "vocabulary" of audio AI)
- These acoustic tokens power modern speech generation (text-to-speech, voice conversion, audio LLMs)
- But what do these tokens actually represent? Content? Speaker identity? Pitch? All entangled together?

Analogy:
"What if we didn't know
what BERT tokens meant?"

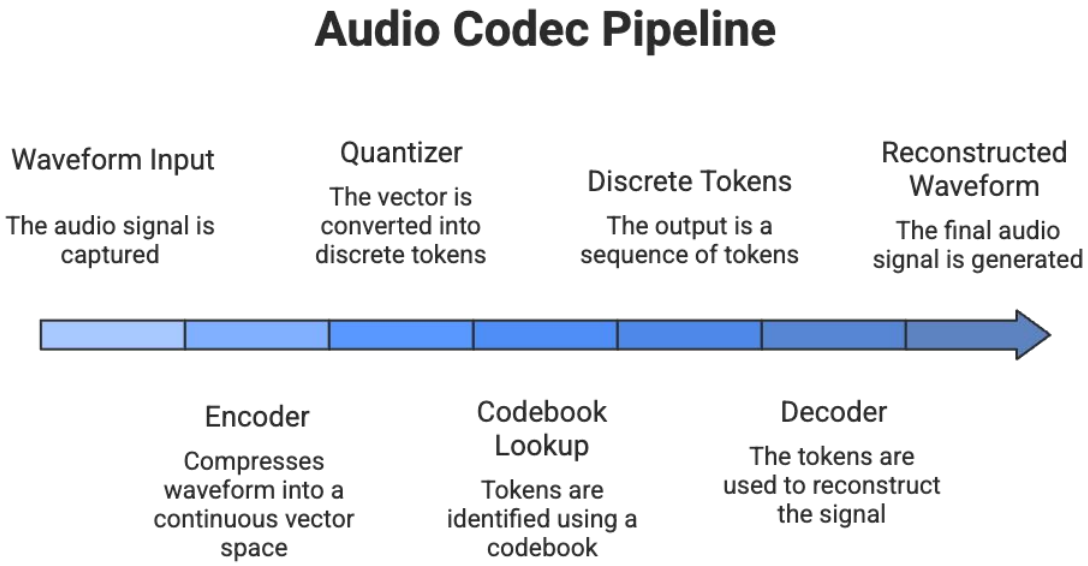
This paper's goal:
Open the black box of
acoustic token representations

Background: Neural Audio Codecs

A neural audio codec compresses continuous audio waveforms into compact sequences of discrete tokens.

- Soundstream (Google, 2021) and Encodec (Meta, 2022) established this compress-then-generate paradigm
- Audio tokens are now the standard building blocks for speech generation with transformer models

"BPE for audio" – converting a continuous signal into a discrete token sequence



Vector Quantization (VQ) & Residual VQ (RVQ)

Vector Quantization (VQ)

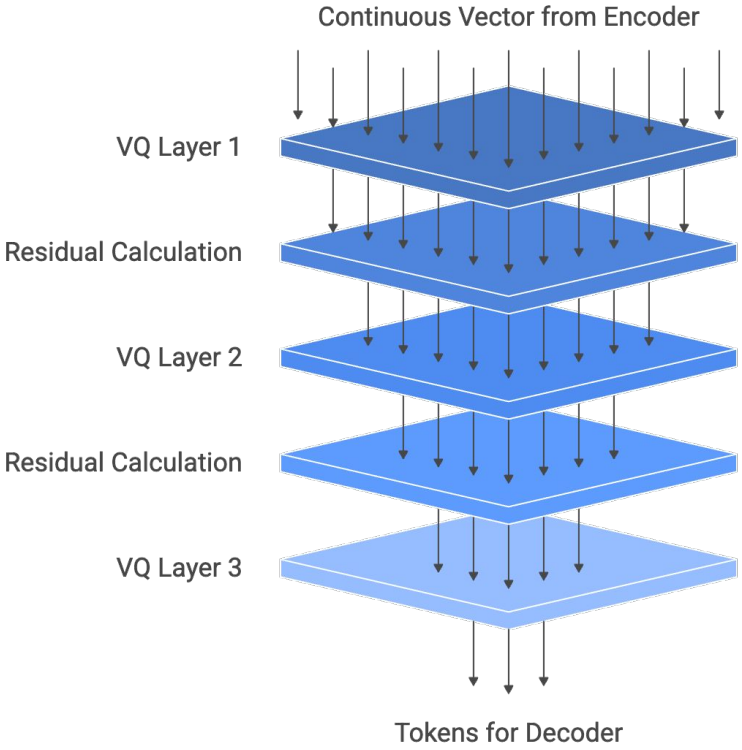
- A codebook of learned prototype vectors (a "dictionary" of audio patterns)
- Map each encoder output to the nearest codebook entry → that index = your token

Residual Vector Quantization (RVQ)

- Stack multiple VQ layers in sequence:
  Layer 1: quantize signal (coarse)
  Layer 2: quantize the residual error
  Layer 3: quantize remaining error
  → Progressively finer detail

Like describing a color:
"blue" → "navy" → "navy with a hint of purple"

Residual Vector Quantization Process



Four Codecs Studied

Descript Audio Codec (DAC)
8-scale RVQ · 1024-entry codebook
First codec to beat Encodec
Factorized + L2-normalized codes

SpeechTokenizer
8-scale RVQ · 1024-entry codebook
HuBERT distillation into RVQ-1
→ Layer 1 encodes linguistic content

Mimi
16-scale split RVQ + Transformer
Causal design · 12.5 Hz · ~1 kbps
Outperforms DAC & SpeechTokenizer

BigCodec
Single VQ · 16384-entry codebook
No RVQ — one large codebook
Similar quality & bitrate as Mimi

HuBERT: a self-supervised speech model (like BERT, but for audio) that learns semantic representations from unlabeled speech via masked prediction

Highlighted codecs are the focus of the paper's synthesis experiments

Distillation: training one model to reproduce another model's learned representations; transfers knowledge without the original model's full complexity

Paper's Two-Stage Approach

Stage 1: Analysis

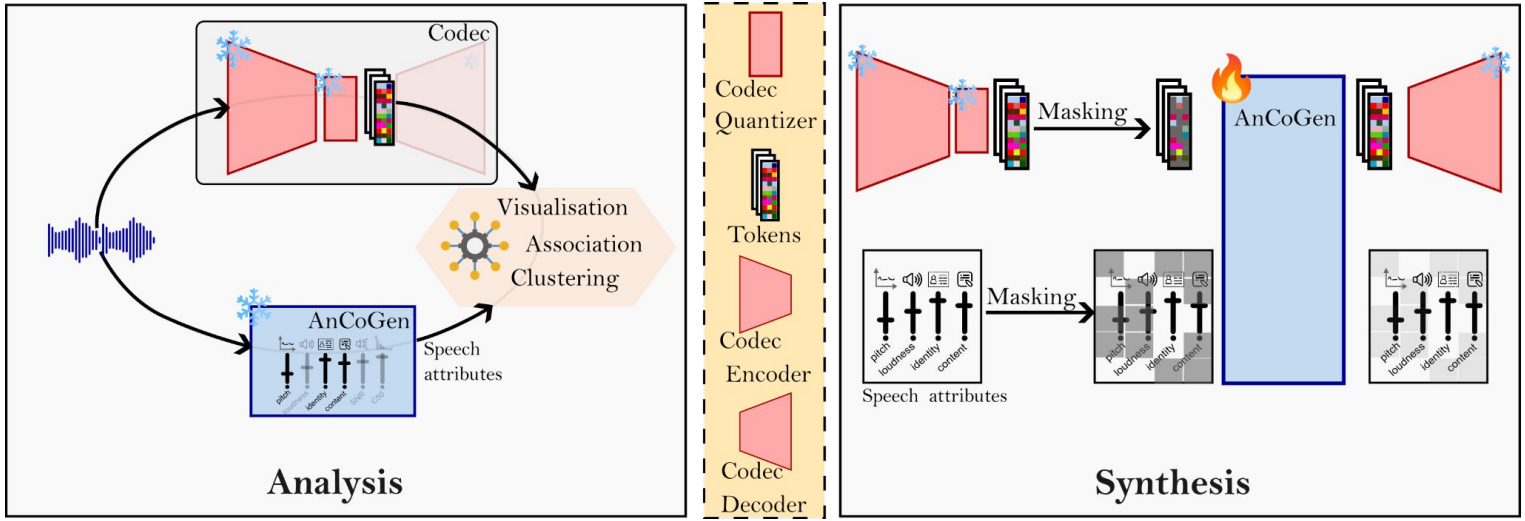
Observe how speech attributes map to codec tokens (no training needed)

- Token-to-HuBERT co-occurrence
- t-SNE embedding visualization
- Mutual information estimation

Stage 2: Synthesis

Train AnCoGen to predict speech attributes from codec tokens and vice versa (bidirectional prediction)

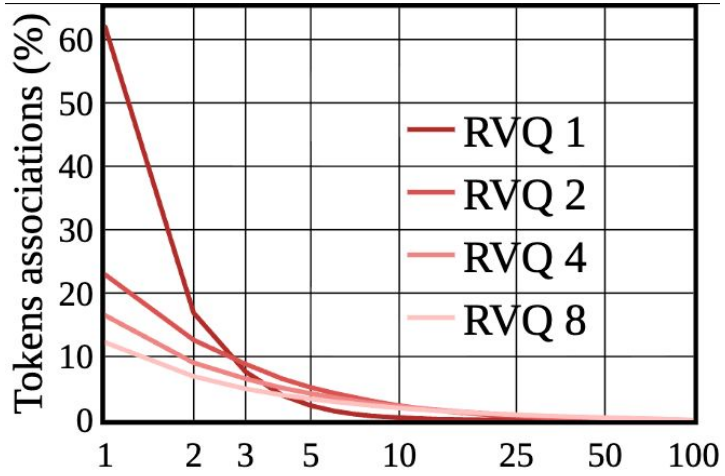
- Encoder-decoder Transformer
- Masked prediction – hide some tokens and predict them from surrounding context
- Cross-entropy loss on masked tokens



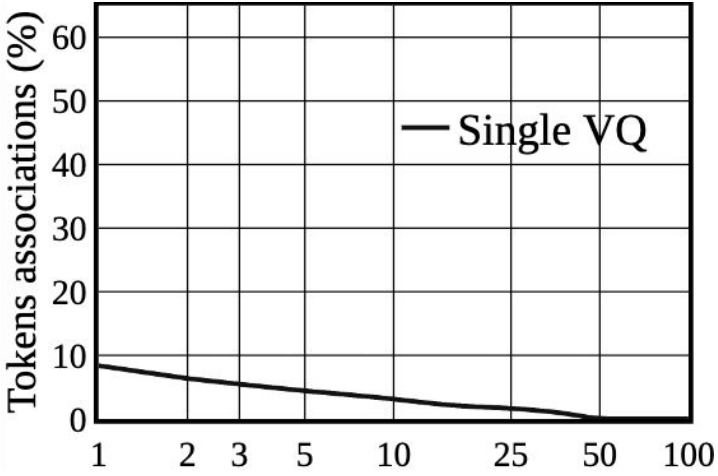
Paper Figure 1: Analysis & Synthesis framework diagram

Analysis: Content – Token Associations

How strongly do codec tokens correspond to semantic (HuBERT) tokens?
Built a co-occurrence matrix on LibriSpeech test-clean for each codec and scale.



2c: SpeechTokenizer



2a: BigCodec

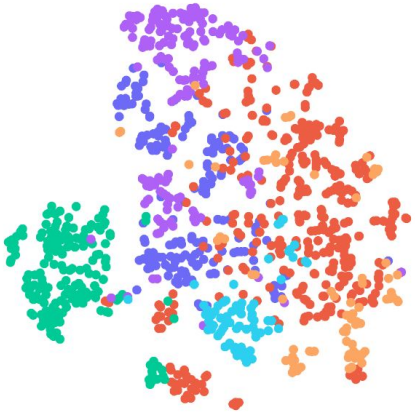
Paper Figure 2: HuBERT/Codecs tokens associations on LibriSpeech test-clean: top-k HuBERT tokens usage per codec token/scale

Key Findings:

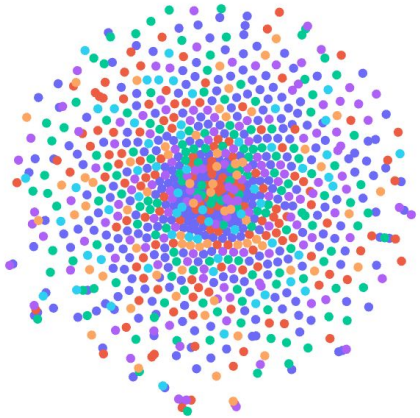
- SpeechTokenizer RVQ-1: ~55% of tokens map to a single top HuBERT token (strong alignment)
- Higher RVQ scales → weaker mapping (content mostly encoded in first scale)
- BigCodec single VQ: only ~8% top-1 alignment (weak – not trained for semantic alignment)

Analysis: Content – Embedding Space (t-SNE)

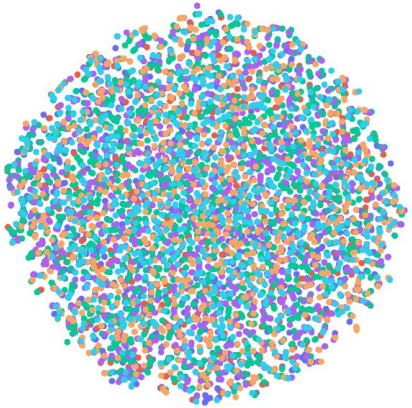
t-SNE (t-distributed Stochastic Neighbor Embedding): a dimensionality reduction technique that maps high-dimensional data to 2D, preserving local neighborhood structure



Paper Figure 3a: Content – SpeechTokenizer RVQ-1



Paper Figure 3c: Content – SpeechTokenizer RVQ-8



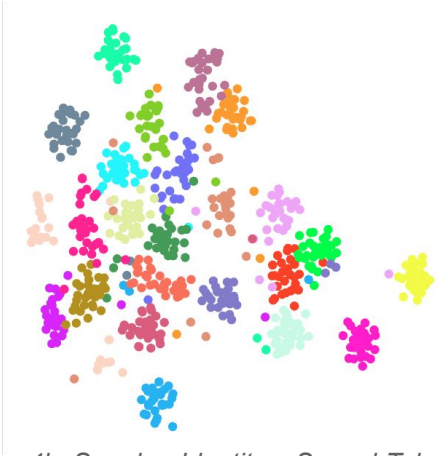
Paper Figure 3d: Content – BigCodec (Single VQ)

- SpeechTokenizer RVQ-1: clear phonetic clusters – vowels, fricatives, stops group together
- SpeechTokenizer RVQ-8: diffuse cloud – no discernible linguistic structure
- BigCodec: some clusters visible, but they don't map cleanly to sound categories

Analysis: Identity & Pitch – Embedding Space (t-SNE)

Speaker Identity

Averaged embeddings per utterance (720 utterances, 24 speakers), colored by speaker.

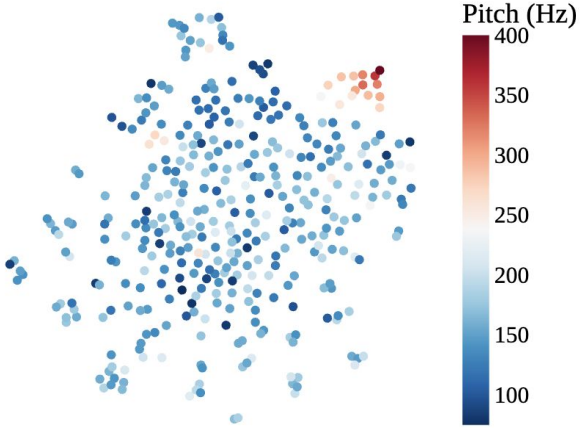


Paper Figure 4b: Speaker Identity – SpeechTokenizer RVQ-4

- RVQ-1: barely present (by design)
- RVQ-4: highly prevalent clusters
- RVQ-8: dispersed but still present

Pitch

Filtered to tokens associated with a vowel HuBERT token, colored by mean pitch (Hz).

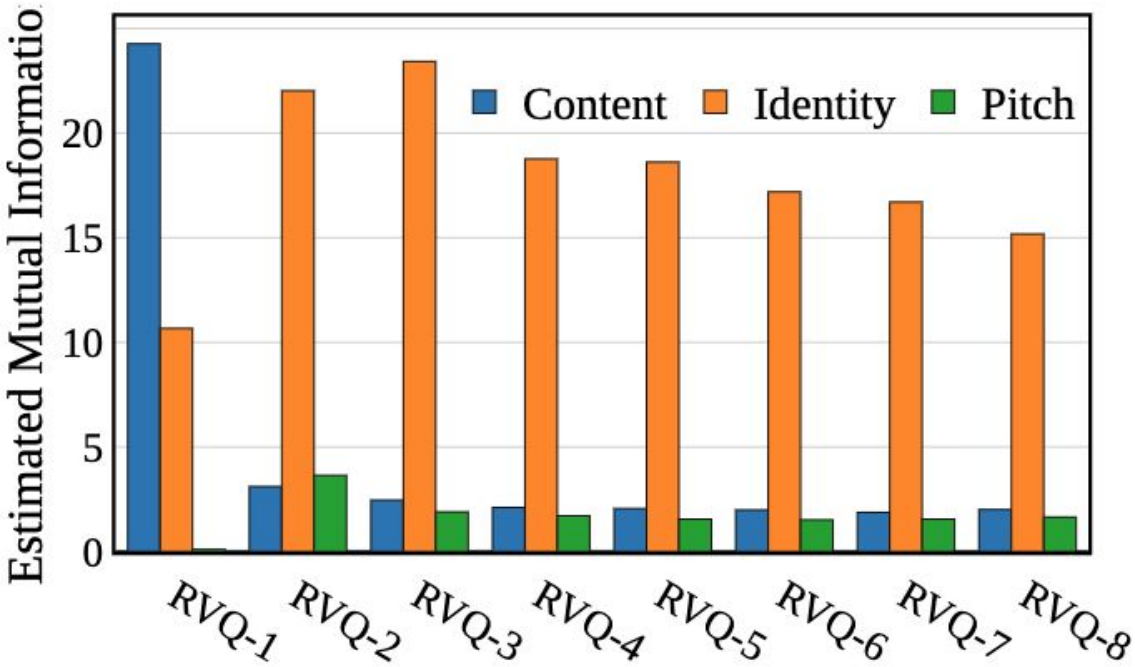


Paper Figure 5b: Pitch – SpeechTokenizer RVQ-2

- Almost no pitch structure visible
- Only small high-pitch cluster on RVQ-2
- Pitch is the hardest attribute to decode

Key insight: content is organized in early RVQ scales, identity in middle scales, but pitch lacks clear structure; it's a perceptual attribute not captured by RVQ refinement

Analysis: Mutual Information



Paper Figure 6: Mutual information between SpeechTokenizer scales and speech attributes

Quantitative analysis of how much information each RVQ scale contains about content, speaker identity, and pitch (SpeechTokenizer, 8 scales).

Key Observations:

- Content MI peaks at RVQ-1 (HuBERT distillation effect)
- Identity MI peaks at RVQ-3/4 (speaker info in middle scales)
- Pitch MI is consistently low across all scales
- Quantitatively confirms the t-SNE visual observations

Synthesis: AnCoGen-Codec Method

AnCoGen = Analysis, Control, and Generation

Architecture: Encoder-decoder Transformer

Training

- Convert speech into two parallel sequences: codec tokens + speech attribute tokens
- Randomly mask tokens from both sequences
- Encoder processes visible tokens via multi-head self-attention
- Decoder predicts masked token indices (cross-entropy loss)

Inference

Analysis direction:
Feed codec tokens, mask ALL attribute tokens → model predicts pitch, content, loudness, identity

Generation direction:
Feed attribute tokens, mask ALL codec tokens → model predicts codec tokens → decode to audio

Four speech attributes:
Linguistic Content (from HuBERT) · Pitch Contour · Loudness · Speaker Identity

Enables: speech analysis (extract attributes from tokens), speech resynthesis (roundtrip tokens ↔ attributes), and voice conversion (swap speaker identity)

Synthesis: Examples

Speech Resynthesis/Reconstruction and Voice Conversion Examples

https://0nutation.github.io/SpeechTokenizer.github.io/

Results

		Analysis		Resynthesis		Control
AnCoGen Model	**Bitrate (kbps) ↓ = more compression**	**Pitch (Hz) Avg. Absolute Error (AAE) ↓**	**Content Accuracy (HuBERT content class label predictions) ↑**	**Perceived Audio Quality (1-5 scale) N-MOS ↑**	**Semantic Similarity (Resynth. Vs Original Text) BertScore ↑**	**Voice Conversion Cosine Similarity (COS) ↑**
Melspectrogram (baseline)	51.2	**4.60**	81.20%	4.26	0.87	0.72
BigCodec	1.04	7.04	73.58%	**4.39**	0.82	0.68
SpeechTokenizer	4.00	6.48	**82.00%**	4.34	**0.88**	**0.74**

Best score in each column is **Bold Teal**, second best is underlined | ↑ = higher is better | ↓ = lower is better

Key Insights:

- **SpeechTokenizer** tops content accuracy (82%), consistent with HuBERT-distilled RVQ-1.
  - It also excels at voice conversion (COS 0.74) and resynthesized semantic similarity (BertScore 0.88)
- **BigCodec** achieves highest perceived audio quality (N-MOS 4.39) despite smallest bitrate – impressive!
  - However, it does not perform as well in other categories
- **Melspectrogram baseline** is still competitive, suggesting codecs have room to improve (especially on pitch!)

Key Takeaways & Limitations

Key Takeaways

1. Speech attributes (content, identity, pitch) are entangled within codec token spaces, even in codecs designed for disentanglement

2. AnCoGen provides a practical post-hoc tool to bridge codec tokens and interpretable speech attributes

3. Pitch is the hardest attribute to decode – perceptual features don't naturally align with RVQ structure

Limitations & Open Questions

1. Synthesis experiments cover only 2 of 4 codecs (SpeechTokenizer & BigCodec)

2. All evaluation on LibriSpeech: clean, read English speech only. How would results change for noisy/conversational/ multilingual audio?

3. No comparison with other interpretability methods (e.g., probing classifiers)

Broader insight: as audio tokens become the "words" of audio AI, the need for interpretability tools will only grow, just as it has for text NLP

Discussion

Questions for the class:

1. NLP tokenizers like BPE are designed around known linguistic units. Should audio codec training explicitly optimize for interpretability, or is post-hoc analysis (like this paper) sufficient?

2. The paper shows speech attributes are entangled in codec tokens. How does this compare to how meaning is distributed across subword tokens in text LLMs? Are there useful parallels or key differences?

3. SpeechTokenizer uses HuBERT distillation to align its first RVQ scale with semantic content. What are the trade-offs of baking in this kind of inductive bias vs. letting the model learn freely?

4. Pitch was the hardest attribute to decode from codec tokens. Why might human perceptual attributes be particularly challenging for these representations, and how could this inform future codec design?